Eugenio D. Macias

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EDUCATION

GPA: 4.0

Brown University, Providence, RI | *Aug 2022 – May 2026*
B.S. Applied Mathematics–Computer Science **(Honors)**, *B.A. Mathematics*

United World College (UWC) – *IB Diploma* Mostar, Bosnia & Herzegovina | *Graduated May 2022*

EXPERIENCE

Brown University, *Karniadakis Lab* – **Undergraduate Research Assistant** Providence, RI | *Jan 2026 – Present*
(Python, PyTorch, PINNs, L-BFGS-B, SSBroyden, Adam) DRP-Cahn-Hilliard-PINN

- Benchmarking quasi-Newton optimizers (SSBroyden, L-BFGS-B) vs. Adam for PINNs on the 2D Cahn-Hilliard 4th-order phase-field PDE; analyzing convergence rates and loss landscape geometry

Citi – **Quantitative Analyst Intern**: *Equity Derivatives* NYC | *Jun 2025 – Aug 2025*

- Automated pricing diagnostics in a Python-C++ interface for exotic derivatives in EMEA, NAM, APAC books.
- Validated live payoff fair values and Greeks via stochastic/local vol MC and PDE solvers; accelerated runtime by 2.5x
- Obtained 1st place (540+ participants) in Citi's firmwide ML Competition and 2nd in internal market-making challenge
- LightGBM+DART+SHAP with Bayesian optimization on 6M+ exotics notes; Swish NNs achieving lowest MSE

Brown University – **Undergraduate Teaching Assistant** Providence, RI | *Aug 2025 – Present*

- *APMA 1200 – Operations Research* (Jan 2026–Present): Markov chains, Poisson processes, Martingales, Brownian motion
- *MATH 100 – Calculus II* (Aug–Dec 2025): integration, sequences & series, Taylor expansions, multivariable calculus

D. E. Shaw & Co. – *Latitude Fellowship* NYC | *Apr 2024 – May 2024*

- Built airline sector revenue forecast via multivariate time-series regression in Python; achieved $R^2 = 0.87$ on test set
- Explored alpha-generating signals and execution latency effects for equities and ETFs

Buddywise AI – *Founder's Associate Intern* Berlin, Germany | *May 2023 – Jul 2023*

- Researched and optimized datasets for ML pipelines in safety-critical computer vision; reduced processing time by 35%
- Secured $50,000 in Mixpanel credits; improved A/B testing and UI KPI visibility by 40%

PROJECTS

Projects

Honors Senior Thesis: ***Weak-Form PDE Discovery for Stochastic Collective Dynamics*** | *Advisor: Prof. Bjorn Sandstede (Python, PyTorch, JAX, SciPy)* wsindy-manifold | Thesis

- Applied **WSINDy** to discover interpretable coupled PDEs for macroscopic density evolution from noisy Vicsek-alignment + Morse attraction–repulsion trajectories; sparsity via MSTLS, integration via ETDRK4
- Benchmarked against equation-free ROMs: FFT-aligned snapshots compressed by POD, evolved in latent space by **MVAR** and **LSTM**; evaluated across 7 dynamical regimes, 4 IC families, and multiple noise levels

DRP – Self-Scaled Quasi-Newton Methods for Physics-Informed Neural Networks *(Python, PyTorch, L-BFGS-B, SSBroyden)* | *Karniadakis Lab* Paper | DRP-Cahn-Hilliard-PINN

- Benchmarked self-scaled quasi-Newton optimizers (SSBroyden variants, L-BFGS-B) vs. Adam for PINNs on the 2D Cahn–Hilliard 4th-order phase-field PDE; best $\varepsilon_{L_2} = 0.258$ on 64^2 grid (SSBroyden2)
- Ran 100+ experiments (53 configurations, 3 seeds each); analyzed convergence rates and loss landscape geometry

RL Research: Curiosity-Driven Transfer in ProcGen *(Python, PyTorch, PPO, ICM, RND, CFN)* | *Advisor: Prof. George Konidaris* TransferRL | Paper

- Factorial sweep over four intrinsic signals (ICM, RND, CFN, NLL-Surprise) augmenting **PPO** with IMPALA-CNN on **ProcGen**; normalization (clip/RMS) and $\eta \in \{0.001 - -0.05\}$ as primary experimental variables
- Zero-shot and fine-tuned transfer; dynamics-aware vs. coverage-only signals, 40% cutoff, VLM-guided (CLIP) reward

Hamiltonian & Symplectic NNs for Charged Particle Dynamics *(PINNs, SympNets, PyTorch)* Paper

- Poisson NNs with PDE residual loss learn Hamiltonian structure in EM fields; sub-0.007 MSE, 2% parameter recovery
- SympNets via Lie-Darboux transforms learn symplectomorphisms from trajectories, preserving phase-space volume

Market-Neutral Long-Short Equity Strategy *(Python, Scikit-Learn, Pandas)* Paper

- PCA factor decomposition on 5,000 equities; OLS idiosyncratic alpha residuals; dollar-neutral long-short top/bottom quintile; Sharpe 2.12 out-of-sample

Research Paper: Fourier Analysis and Laplace's Equation *(LaTeX)* Paper

- Fourier series for Laplace's 2D BVP; four Dirichlet configs; L^2 convergence & truncation error analysis (50 pp.)

AWARDS

- **Shelby Davis Scholar**: Full tuition, Brown
- **Sprint Award**: $7,000 research internship grant
- **Citadel** Invitational Terminal 2024
- **Point72 & Correlation1**: Data Scholar 2024
- **Sigma Xi**: Invited Member, Research Honor Society
- **Bridgewater Associates**: Investment Immersion 2024
- **PUTNAM 2024**: top 20% nationwide
- **IYMC**: Silver Medalist, Mathematics

SKILLS & INTERESTS

Math Notes

Skills: Rapid quantitative intuition & mental math; bullet chess (**2070** ELO); AI-accelerated research; socialization
Languages: Spanish (native), English (native), German (B1), French (A2)
Interests: Classical Guitar (**3$^{\mathbf{rd}}$** National, Mexico), Running (**5** marathons), Travel (**45** countries), Asian food